

1.1. Introduction:

In this study, we address the problem of **automated meeting transcript generation and documentation**, where raw meeting audio must be converted into an accurate and usable record of the discussion. The primary objective is to reliably recover the content of the meeting despite transcription errors, conversational speech, multiple speakers, and technical terminology. Following this, the system refines the transcript and generates structured documentation which contains the key discussion points, decisions, and action items, while preserving the factual information present in the original meeting and avoiding unsupported assumptions. The system supports both English and Hindi meeting recordings, with the Hindi pipeline producing an English transcript before the downstream refinement and documentation stages.

1.2. Literature Review:

1.2.1. Rennard et al. [1]

Rennard et al. [1] present a comprehensive survey of **abstractive meeting summarization**, highlighting challenges such as noisy and disfluent speech, multiple speakers, long conversations, and maintaining factual consistency. They organize existing approaches using an **Interpretation–Transformation–Generation** framework, in which the conversation is first interpreted, then transformed into a structured representation, and finally converted into a summary. The survey emphasizes the importance of speaker information, topic segmentation, and conversational context for producing reliable meeting summaries, particularly when identifying decisions and other important information. These observations motivate our use of an intermediate **Transformation layer**, which organizes the refined transcript into evidence-linked topics, discussion units, decisions, and action items before generating the final meeting documentation.

1.2.2. Du et al. [2]

Du et al. [2] investigate **speaker diarization for multi-party meetings**, with particular emphasis on the challenging case of **overlapping speech**, where multiple speakers talk simultaneously. They propose **Speaker Overlap-aware Neural Diarization (SOND)**, which explicitly models overlapping speaker activity and speaker dependencies to improve speaker attribution. Our system addresses the same speaker-attribution challenge using **PyAnnote Community-1** for speaker diarization, followed by alignment with timestamped ASR output.

1.2.3. Jin et al. [3]

Jin et al. [3] propose **CFAS (Consensus-Focused Abstractive Summarization)**, a framework that explicitly considers **topics and consensus points** when generating meeting summaries. Their approach incorporates topic segmentation and consensus identification before performing consensus-aware summarization, giving greater importance to utterances that represent agreements or decisions. Inspired by this work, we extended our Transformation stage with **consensus analysis**, allowing extracted decisions to be evaluated against supporting conversational evidence. The system distinguishes confirmed agreements from proposals, contested discussions, and unresolved decisions rather than treating every decision-like statement as a final decision.

1.2.4. Inoue et al. [4]

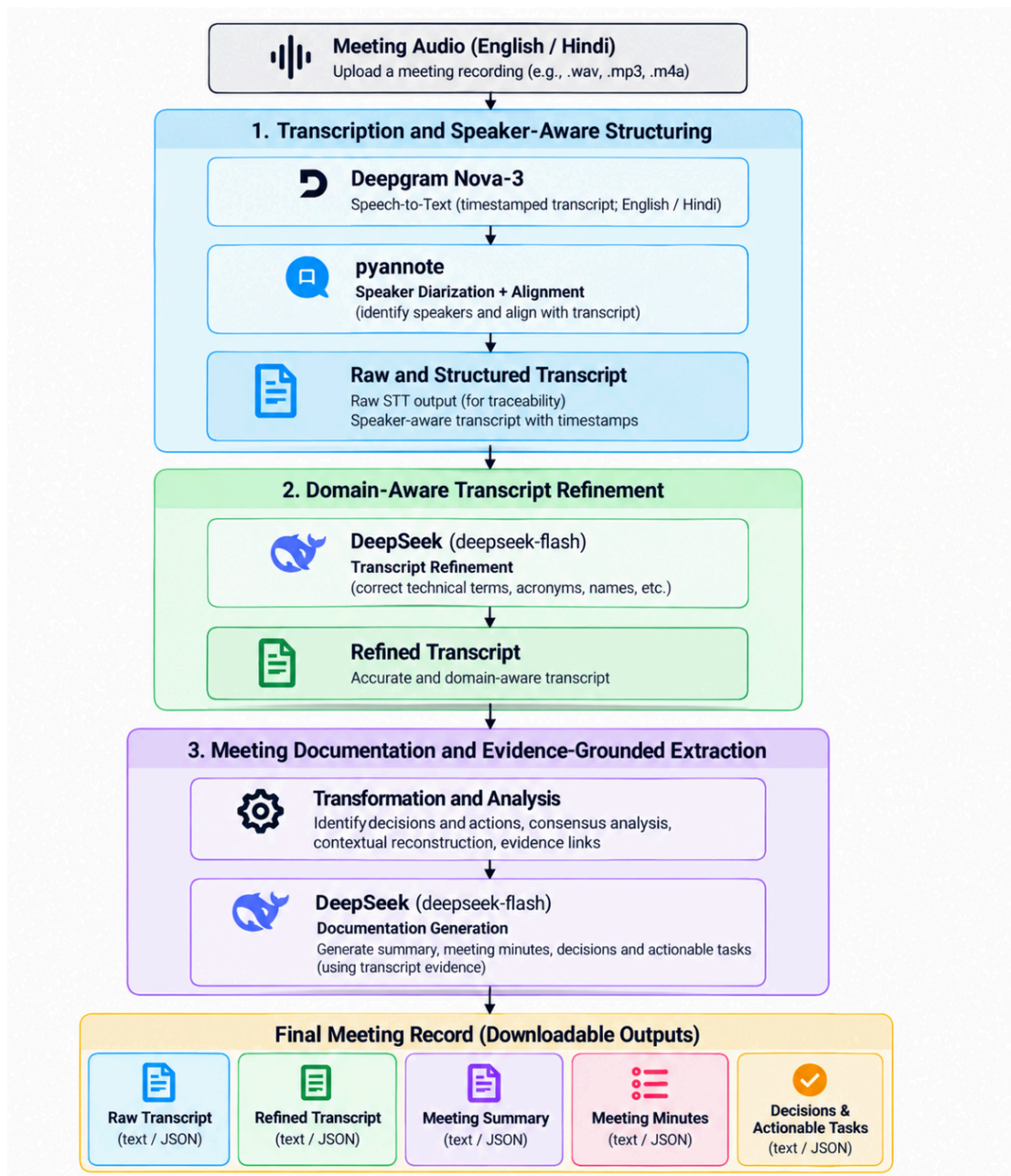
Inoue et al. [4] introduce the **Meeting Decision Tracker**, which identifies decision-related utterances and rewrites them into **self-contained decision statements**. Their work addresses the problem that spoken decisions are often context-dependent, with important information omitted or represented through references that require surrounding dialogue for interpretation. By incorporating conversational context, their system reconstructs these utterances into clearer written decisions. Inspired by this approach, we added contextual reconstruction to our Transformation stage, using surrounding transcript segments to make extracted decisions self-contained while retaining links to the original evidence.

REFERENCES:

- [1] V. Rennard, G. Shang, J. Hunter, and M. Vazirgiannis, "Abstractive Meeting Summarization: A Survey," *Transactions of the Association for Computational Linguistics*, vol. 11, 2023. [TransACL](#)
- [2] Z. Du, S. Zhang, S. Zheng, and Z.-J. Yan, "Speaker Overlap-aware Neural Diarization for Multi-party Meeting Analysis," *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 7458–7469, 2022. [ACL Anthology](#)
- [3] Y. Jin, Q. Shi, and Q. Liu, "CFAS: Consensus-Focused Abstractive Meeting Summarization through Multi-Party Discourse Modeling," *Journal of King Saud University Computer and Information Sciences*, vol. 37, art. 187, 2025. [Springer](#)
- [4] S. Inoue, H. Nguyen, H. Pham, T. Liu, and M.-T. Nguyen, "Meeting Decision Tracker: Making Meeting Minutes with De-Contextualized Utterances," *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing: System Demonstrations*, pp. 88–94, 2022.

1.3 Methodology

The proposed system processes a meeting recording through a sequential multimodel pipeline comprising **speech transcription and speaker diarization, transcript refinement, evidence-based processing, and meeting documentation**. The system supports both **English and Hindi meeting recordings**, while maintaining the same downstream refinement and documentation stages. Each stage operates on the output of the preceding stage, with intermediate results retained to maintain traceability throughout the pipeline.



1.3.1 Transcription and Speaker-Aware Structuring

The first stage converts the meeting recording into a timestamped transcript using **Deepgram Nova-3** for automatic speech recognition. The system supports both **English and Hindi meeting recordings**, allowing the subsequent processing stages to operate on transcripts from either language. The raw ASR output is retained separately and serves as the reference for all subsequent processing.

To recover the conversational structure, **PyAnnote Community-1** is used for speaker diarization. The diarization output is aligned with the timestamped ASR segments, associating each segment with its corresponding speaker and temporal boundaries. The resulting representation contains the speaker identity, start time, end time, and transcript text required by the downstream stages.

The raw transcript is preserved without modification, while the aligned speaker-aware representation is used for subsequent processing. This separation provides traceability between the original ASR output and any corrections introduced later in the pipeline.

1.3.2 Domain-Aware Transcript Refinement

The second stage uses **DeepSeek *deepseek-flash*** to refine the speaker-aware transcript. This model is specifically responsible for correcting likely ASR errors rather than summarizing or interpreting the meeting. Particular attention is given to technical terminology, acronyms, names, identifiers, and other domain-specific terms that are prone to recognition errors.

The refinement process is deliberately conservative. Speaker attribution, timestamps, ordering, numerical values, negation, uncertainty, questions, proposals, and commitments are preserved while correcting the textual representation of the speech. The model produces the refined transcript together with structured information describing the changes made to individual segments.

A deterministic validation layer is applied to the model output before the changes are accepted. It compares proposed modifications against the corresponding raw segments and checks for unsafe edits, including unsupported modifications and changes to critical information such as numbers or negation. This keeps the refined transcript grounded in the original ASR output while allowing genuine recognition errors to be corrected.

1.3.3 Meeting Documentation and Evidence-Grounded Extraction

The final stage converts the refined transcript into a structured meeting record.

Before the documentation model generates the final output, an intermediate **Transformation layer** processes the transcript to identify candidate decisions and action items and associates them with the relevant transcript segment IDs. The Transformation layer also performs **consensus analysis** to distinguish confirmed agreements from proposals, tentative statements, contested points, and unresolved decisions. This prevents statements discussing a possible course of action from being incorrectly represented as finalized decisions. Where the meaning of a decision depends on surrounding conversation, **contextual reconstruction** is used to produce a self-contained representation while retaining references to the original transcript evidence.

The resulting evidence-linked information, together with the refined transcript, is passed to a second **DeepSeek deepseek-flash** stage. This model has a distinct role from the refinement model: it generates the final meeting documentation, including the **summary, organized minutes, decisions, and actionable tasks**.

The documentation stage follows an evidence-grounded approach. **Owners and deadlines are included only when they are explicitly supported by the meeting**. Proposals, recommendations, conditional statements, and unresolved discussions are not promoted to confirmed decisions or assignments. This ensures that the generated documentation reflects what was actually established during the meeting rather than filling gaps through inference.

The final record is stored in both **human-readable and machine-readable formats**. Decisions and action items retain references to their supporting transcript segments, allowing the user to inspect the relevant speaker, timestamp, and transcript content. This provides a traceable path from the original recording through transcription and refinement to the final meeting documentation.

1.3. Interactive Application:

- **Audio Upload:**

The application provides a single interface for uploading a meeting recording in English or Hindi and initiating the complete processing pipeline. The uploaded recording is passed through transcription, speaker diarization, refinement, evidence processing, and documentation without requiring the user to execute individual pipeline components.

- **Pipeline Status:**

Processing progress is displayed stage-by-stage, covering transcription, transcript refinement, transformation, and documentation generation. The user can see whether each stage is processing, completed, or failed.

- **Transcript Review:**

The application provides access to both the **raw transcript** generated by the STT system and the **refined transcript** produced after domain-aware correction. This allows the user to compare the two versions.

- **Meeting Documentation:**

The final interface presents the generated **summary, meeting minutes, decisions, and actionable tasks** in an organized format.

- **Evidence-Based Review:**

Decisions and action items can be traced back to their supporting transcript segments. The application displays the relevant speaker, timestamp, and transcript content, allowing the user to verify the extracted information.

- **Warnings and Uncertainty:**

Items with ambiguous or weaker supporting evidence are flagged for review instead of being presented as fully certain.

- **Downloadable Outputs:**

The application provides downloadable outputs for the different stages, including:

- Raw transcript (English/Hindi)
- Refined transcript
- Meeting summary
- Meeting minutes
- Decisions
- Actionable tasks
- Machine-readable structured meeting record

- **Human- and Machine-Readable Results:**

The final meeting record is available in both readable text formats and structured formats suitable for further processing.

- **Error Handling:**

Unsupported, empty, unreadable, or unsuccessfully processed recordings are

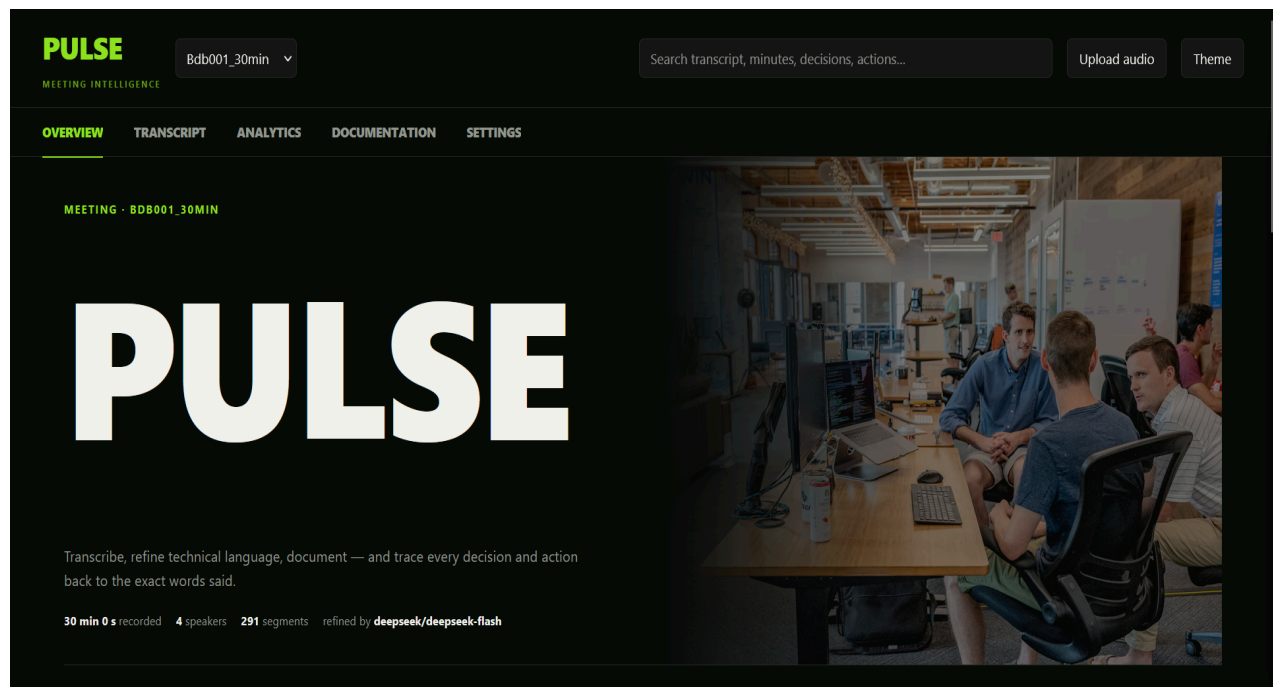
reported through clear error messages in the interface rather than exposing internal system errors.

- **Single End-to-End Workflow:**

The application connects all processing stages into one workflow, allowing a new recording to be uploaded and processed through to the final downloadable meeting record without manually running individual pipeline components.

- **Ask PULSE:** The application includes an interactive **Ask PULSE** feature that allows users to query the processed meeting record using natural-language questions. Responses are grounded in the transcript and documented meeting evidence, enabling users to quickly retrieve relevant decisions, discussions, and action items without manually searching through the entire transcript.

1.5 Application Interface



PULSE

MEETING INTELLIGENCE

test2.wav ▾

Search transcript, minutes, decisions, actions...

Upload audio

Theme

OVERVIEW

TRANSCRIPT

ANALYTICS

DOCUMENTATION

SETTINGS

Processing Bdb001_30min

LIVE STATUS FROM THE PIPELINE

1. STT done · 110.5s	2. Diarization done · 116.9s	3. Alignment done · 0.1s	4. LLM1 Refinement done · 81.3s	5. Transformation running	6. LLM2 Documentation pending
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PULSE - interaction patterns (TextRoll) adapted from Skiper UI — skiperui.com

PULSE

MEETING INTELLIGENCE

Bdb001_30min ▾

Search transcript, minutes, decisions, actions...

Upload audio

Theme

OVERVIEW

TRANSCRIPT

ANALYTICS

DOCUMENTATION

SETTINGS

Meeting timeline

0:00 ~ 29:59.5 · CLICK A BLOCK TO OPEN IT

Who spoke

Cited by documentation

Refined by LLM1

■ Speaker 00 ■ Speaker 01 ■ Speaker 02 ■ Speaker 03

Refinement log

15 CHANGES · 6 REJECTED BY VALIDATION

SEG	ORIGINAL	REFINED	REASON (FROM LLM1)	CONF.
#87	Kepstrol	MFCC	Misrecognized technical term; context of speech features suggests MFCC (Mel-frequency cepstral coefficients).	0.9
#87	plp	PLP	Capitalization of acronym for perceptual linear prediction.	0.9
#97	and		Removed STT duplicate word	0.9

PULSE

MEETING INTELLIGENCE

test2

Search transcript, minutes, decisions, actions...

Upload audio

Ask PULSE

Theme

OVERVIEWTRANSCRIPTANALYTICSDOCUMENTATIONSETTINGS

Documentation

SUMMARYMINUTESDECISIONS (active)ACTIONSWARNINGS

Summary (.txt)Minutes (.txt)Decisions (.txt)Action Items (.txt)Final Record (.json)

DECISION

status: not recorded by pipeline

validation warning

PostgreSQL remains the source of truth for the ledger; Kafka events are transport and workflow coordination, not the authoritative balance store.

segments 51, 52, 59 · Speaker 03, Speaker 00 · 20:05.3–23:41.7

DECISION

status: not recorded by pipeline

validation warning

Istio is the service mesh currently in the staging cluster and is staying for now; no replacement is being evaluated during this migration.

segments 47, 48 · Speaker 03, Speaker 00 · 18:28.8–19:17.6

DECISION

Close

PostgreSQL remains the source of truth for the ledger; Kafka events are transport and workflow coordination, not the authoritative balance store.

Source segments

51, 52, 59

Speaker(s)

Speaker 03, Speaker 00

Time span

20:05.3 – 23:41.7

Status

not recorded by pipeline

validation

cited segments mention pending/unapproved status; verify

Consensus: HIGH

confirmed

internal heuristic score 0.85 - not a calibrated probability

Reconstructed (model): The team confirmed that PostgreSQL remains the authoritative ledger store, while Kafka events are transport and workflow coordination, not the authoritative balance store.

Why?

Decision statement: #51 "We should record that the current architecture decision is to keep PostgreSQL as the source of truth"

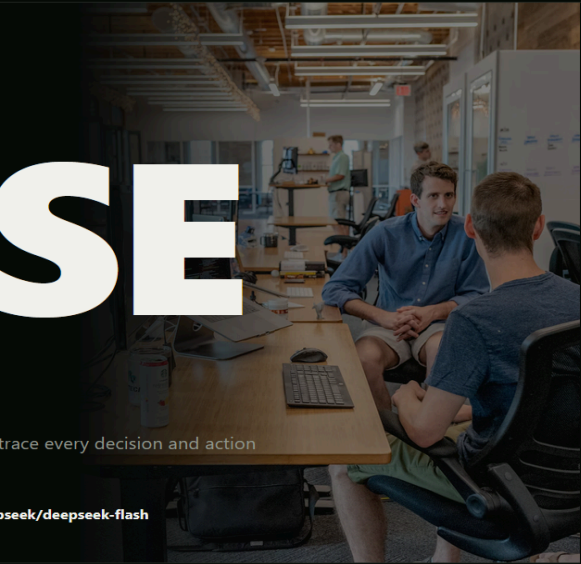
Search transcript, minutes, decisions, actions...

Upload audio

Ask PULSE

Theme

DOCUMENTATION (active)SETTINGS



SE

Trace every decision and action

Deepseek/deepseek-flash

Ask PULSE

Close

Answers grounded in this meeting · based on transcript & documentation

YOU ASKED

is there any mention about kubernetes?

Yes. The deployment target is Kubernetes on three availability zones, with namespace payments-prod and service account payment-orchestrator.

confidence: high

check the evidence below — answers can be wrong

EVIDENCE (1)

Segment 8 · Speaker 00 · 3:04.5–3:29.6

The deployment target is Kubernetes on three availability zones. The namespace is payments-prod, and the service account is payment-orchestrator.

Original STT: "The deployment target is kube net ease on three availability zones. The namespace is payments-prod, and the service account is payment-orchestrator."

e.g. What did we decide about PostgreSQL?

Ask

Enter to ask · Shift+Enter for a new line